```
Bắt đầu lập trình hoặc tạo mã bằng trí tuệ nhân tạo (AI).
```

Bài giảng: Giá trị riêng, vector riêng (Eigenvalue, Eigenvector)

```
import matplotlib.pyplot as plt
import matplotlib as mpl
import numpy as np
import sympy as sy
sy.init_printing()
plt.style.use('ggplot')
```

```
np.set_printoptions(precision=3)
np.set_printoptions(suppress=True)
```

Giá trị riêng, vector riêng (Eigenvalue, Eigenvector)

An *eigenvector* of an $n \times n$ matrix A is a nonzero vector x such that $Ax = \lambda x$ for some scalar λ. A scalar λ is called an *eigenvalue* of A if there is a nontrivial solution x of $Ax = \lambda x$, such an x is called an eigenvector corresponding to λ.

Rewrite the equation,

$$(A - \lambda I)x = 0$$

Since the eigenvector should be a nonzero vector, which means:

1. The column or rows of $(A - \lambda I)$ are linearly dependent
2. $(A - \lambda I)$ is not full rank, $Rank(A) < n$.
3. $(A - \lambda I)$ is not invertible.
4. $\det(A - \lambda I) = 0$, which is called *characteristic equation*.

Consider a matrix A

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 1 \\ 2 & -2 & 3 \end{bmatrix}$$

Set up the characteristic equation,

$$\det\left(\begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 1 \\ 2 & -2 & 3 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}\right) = 0$$

Use SymPy `charpoly` and `factor`, we can have straightforward solutions for eigenvalues.

```
lamda = sy.symbols('lamda') # 'lamda' withtout 'b' is reserved for SymPy, lambda is reserved for Python
```

`charpoly` returns characteristic equation.

```
A = sy.Matrix([[1, 0, 0], [1, 0, 1], [2, -2, 3]])
p = A.charpoly(lamda); p
```

$$\text{PurePoly}\left(\lambda^3 - 4\lambda^2 + 5\lambda - 2, \lambda, domain = \mathbb{Z}\right)$$

Factor the polynomial such that we can see the solution.

```
sy.factor(p)
```

$$\text{PurePoly}\left(\lambda^3 - 4\lambda^2 + 5\lambda - 2, \lambda, domain = \mathbb{Z}\right)$$

From the factored characteristic polynomial, we get the eigenvalue, and $\lambda = 1$ has algebraic multiplicity of 2, because there are two $(\lambda - 1)$. If not factored, we can use `solve` instead.

```
sy.solve(p,lamda)
```

$[1,\ 2]$

Or use `eigenvals` directly.

```
A.eigenvals()
```

$\{1 : 2,\ 2 : 1\}$

To find the eigenvector corresponding to λ, we substitute the eigenvalues back into $(A - \lambda I)x = 0$ and solve it. Construct augmented matrix with $\lambda = 1$ and perform rref.

```
(A - 1*sy.eye(3)).row_join(sy.zeros(3,1)).rref()
```

$$\left(\begin{bmatrix}1 & -1 & 1 & 0\\ 0 & 0 & 0 & 0\\ 0 & 0 & 0 & 0\end{bmatrix},\ (0,)\right)$$

The null space is the solution set of the linear system.

$$\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix} = \begin{bmatrix}x_2 - x_3\\ x_2\\ x_3\end{bmatrix} = x_2\begin{bmatrix}1\\ 1\\ 0\end{bmatrix} + x_3\begin{bmatrix}-1\\ 0\\ 1\end{bmatrix}$$

This is called *eigenspace* for $\lambda = 1$, which is a subspace in $\mathbb{R}^3$. All eigenvectors are inside the eigenspace.

We can proceed with $\lambda = 2$ as well.

```
(A - 2*sy.eye(3)).row_join(sy.zeros(3,1)).rref()
```

$$\left(\begin{bmatrix}1 & 0 & 0 & 0\\ 0 & 1 & -\frac{1}{2} & 0\\ 0 & 0 & 0 & 0\end{bmatrix},\ (0,\ 1)\right)$$

The null space is the solution set of the linear system.

$$\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix} = \begin{bmatrix}0\\ \frac{1}{2}x_3\\ x_3\end{bmatrix} = x_3\begin{bmatrix}0\\ \frac{1}{2}\\ 1\end{bmatrix}$$

To avoid troubles of solving back and forth, SymPy has `eigenvects` to calcuate eigenvalues and eigenspaces (basis of eigenspace).

```
eig = A.eigenvects(); eig
```

$$\left[\left(1,\ 2,\ \left[\begin{bmatrix}1\\ 1\\ 0\end{bmatrix},\ \begin{bmatrix}-1\\ 0\\ 1\end{bmatrix}\right]\right),\ \left(2,\ 1,\ \left[\begin{bmatrix}0\\ \frac{1}{2}\\ 1\end{bmatrix}\right]\right)\right]$$

To clarify what we just get, write

```
print('Eigenvalue = {0}, Multiplicity = {1}, Eigenspace = {2}'.format(eig[0][0], eig[0][1], eig[0][2]))
```

```
Eigenvalue = 1, Multiplicity = 2, Eigenspace = [Matrix([
[1],
[1],
[0]]), Matrix([
[-1],
[ 0],
[ 1]])]
```

```
print('Eigenvalue = {0}, Multiplicity = {1}, Eigenspace = {2}'.format(eig[1][0], eig[1][1], eig[1][2]))
```

```
Eigenvalue = 2, Multiplicity = 1, Eigenspace = [Matrix([
[  0],
[1/2],
[  1]])]
```

NumPy Functions for Eigenvalues and Eigenspace

Convert SymPy matrix into NumPy float array.

```
A = np.array(A).astype(float); A
```

```
array([[ 1.,  0.,  0.],
       [ 1.,  0.,  1.],
       [ 2., -2.,  3.]])
```

`.eigvals()` and `.eig(A)` are handy functions for eigenvalues and eigenvectors.

```
np.linalg.eigvals(A)
```

```
array([2., 1., 1.])
```

```
np.linalg.eig(A) #return both eigenvalues and eigenvectors
```

```
EigResult(eigenvalues=array([2., 1., 1.]), eigenvectors=array([[ 0.   ,  0.   ,  0.408],
       [ 0.447,  0.707, -0.408],
       [ 0.894,  0.707, -0.816]]))
```

An Example

Consider a matrix A

```
A = sy.Matrix([[-4, -4, 20, -8, -1],
               [14, 12, 46, 18, 2],
               [6, 4, -18, 8, 1],
               [11, 7, -37, 17, 2],
               [18, 12, -60, 24, 5]])
A
```

$$\begin{bmatrix} -4 & -4 & 20 & -8 & -1 \\ 14 & 12 & 46 & 18 & 2 \\ 6 & 4 & -18 & 8 & 1 \\ 11 & 7 & -37 & 17 & 2 \\ 18 & 12 & -60 & 24 & 5 \end{bmatrix}$$

Find eigenvalues.

```
eig = A.eigenvals()
eig
```

$$\left\{2 : 2,\ 3 : 1,\ \frac{5}{2} - \frac{\sqrt{1473}}{2} : 1,\ \frac{5}{2} + \frac{\sqrt{1473}}{2} : 1\right\}$$

Or use NumPy functions, show the eigenvalues.

```
A = np.array(A)
A = A.astype(float)
eigval, eigvec = np.linalg.eig(A)
eigval
```

```
array([ 21.69, -16.69,   3.  ,   2.  ,   2.  ])
```

And corresponding eigenvectors.

```
eigvec
```

```
array([[-0.124, -0.224,  0.   , -0.039,  0.611],
       [ 0.886, -0.543, -0.894, -0.216, -0.149],
       [ 0.124,  0.224,  0.   ,  0.   , -0.   ],
       [ 0.216,  0.392,  0.447,  0.255, -0.462],
       [ 0.371,  0.672, -0.   , -0.942,  0.625]])
```

A Visualization Example

Let

$$A = \begin{bmatrix} 1 & 6 \\ 5 & 2 \end{bmatrix}$$

find the eigenvalues and vectors, then visualize in $\mathbb{R}^2$

Use characteristic equation $|A - \lambda I| = 0$

$$\left| \begin{bmatrix} 1 & 6 \\ 5 & 2 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} \right| = 0$$

```
lamda = sy.symbols('lamda')
A = sy.Matrix([[1,6],[5,2]])
I = sy.eye(2)
```

```
A - lamda*I
```

$$\begin{bmatrix} 1 - \lambda & 6 \\ 5 & 2 - \lambda \end{bmatrix}$$

```
p = A.charpoly(lamda);p
```

$$\text{PurePoly}\left(\lambda^2 - 3\lambda - 28, \lambda, domain = \mathbb{Z}\right)$$

```
sy.factor(p)
```

$$\text{PurePoly}\left(\lambda^2 - 3\lambda - 28, \lambda, domain = \mathbb{Z}\right)$$

There are two eigenvalues: 7 and 4. Next we calculate eigenvectors.

```
(A - 7*sy.eye(2)).row_join(sy.zeros(2,1)).rref()
```

$$\left(\begin{bmatrix} 1 & -1 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \ (0,) \right)$$

The eigenspace for $\lambda = 7$ is

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = x_2 \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Any vector is eigenspace as long as $x \neq 0$ is an eigenvector. Let's find out eigenspace for $\lambda = 4$.

```
(A + 4*sy.eye(2)).row_join(sy.zeros(2,1)).rref()
```

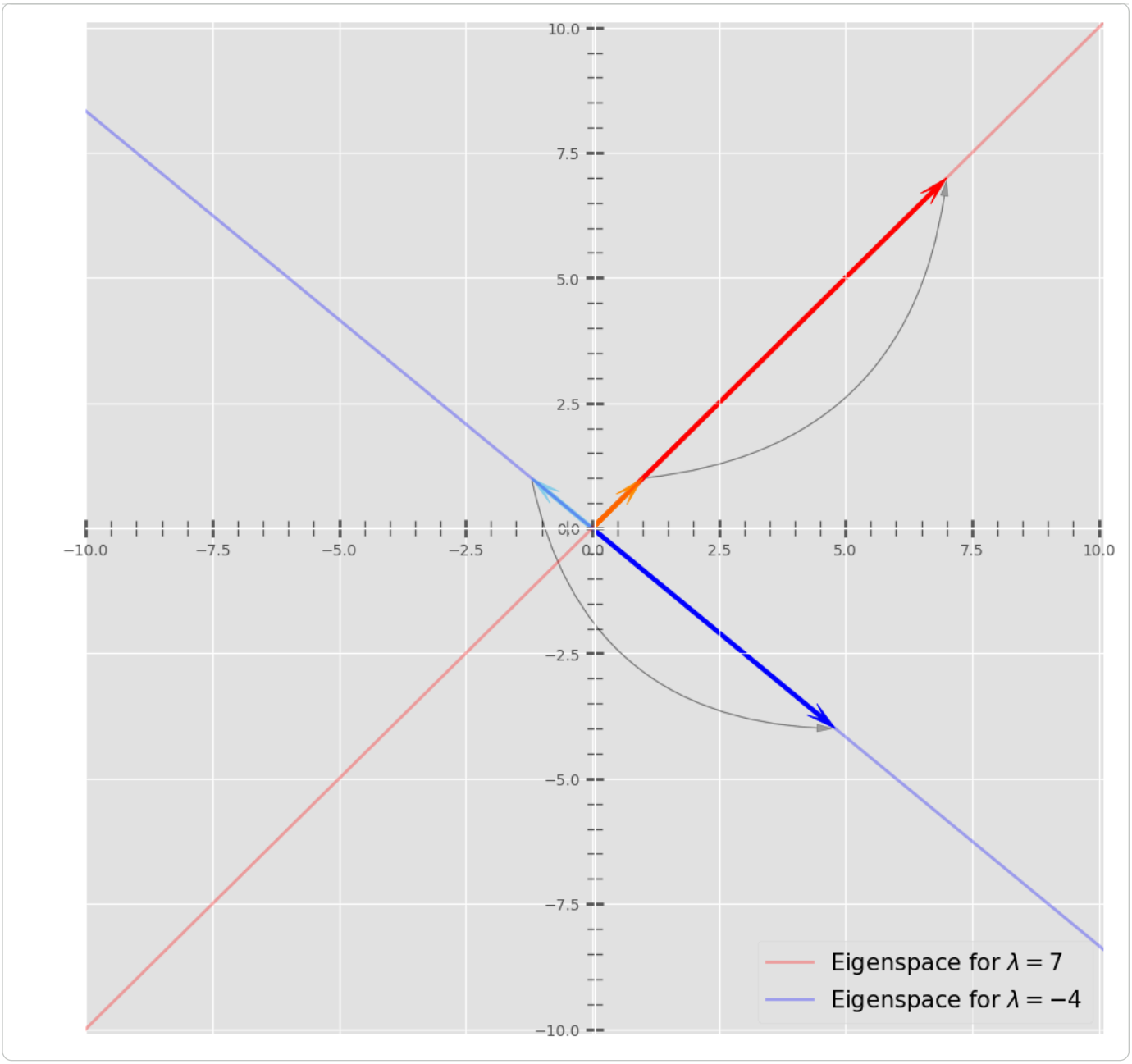
$$\left(\begin{bmatrix} 1 & \frac{6}{5} & 0 \\ 0 & 0 & 0 \end{bmatrix}, \ (0,) \right)$$

The eigenspace for $\lambda = -4$ is

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = x_2 \begin{bmatrix} -\frac{6}{5} \\ 1 \end{bmatrix}$$

Let's plot both eigenvectors as $(1, 1)$ and $(-6/5, 1)$ and multiples with eigenvalues.

```
import numpy as np
import matplotlib.pyplot as plt
import matplotlib as mpl

fig, ax = plt.subplots(figsize=(12, 12))
arrows = np.array([ [[0, 0, 1, 1]],
                    [[0, 0, -6/5, 1]],
                    [[0, 0, 7, 7]],
                    [[0, 0, 24/5, -4]] ])
colors = ['darkorange', 'skyblue', 'r', 'b']

for i in range(arrows.shape[0]):
    X, Y, U, V = zip(*arrows[i, :, :])
    ax.arrow(X[0], Y[0], U[0], V[0], color=colors[i], width=0.08,
             length_includes_head=True, head_width=0.3, head_length=0.6,
             overhang=0.4, zorder=-i)

# Eigenspace for λ = 7
x = np.arange(-10, 10.6, 0.5)
y = x
ax.plot(x, y, lw=2, color='red', alpha=0.3, label=r'Eigenspace for $\lambda = 7$')

# Eigenspace for λ = -4
x = np.arange(-10, 10.6, 0.5)
y = -5/6 * x
ax.plot(x, y, lw=2, color='blue', alpha=0.3, label=r'Eigenspace for $\lambda = -4$')

# Annotation Arrows
style = "Simple, tail_width=0.5, head_width=5, head_length=10"
kw = dict(arrowstyle=style, color="k")

a = mpl.patches.FancyArrowPatch((1, 1), (7, 7), connectionstyle="arc3,rad=.4", **kw, alpha=0.3)
plt.gca().add_patch(a)

a = mpl.patches.FancyArrowPatch((-6/5, 1), (24/5, -4), connectionstyle="arc3,rad=.4", **kw, alpha=0.3)
plt.gca().add_patch(a)

# Legend
leg = ax.legend(fontsize=15, loc='lower right')
leg.get_frame().set_alpha(0.5)

# Axis, Spines, Ticks
ax.axis([-10, 10.1, -10.1, 10.1])
ax.spines['left'].set_position('center')
ax.spines['right'].set_color('none')
ax.spines['bottom'].set_position('center')
ax.spines['top'].set_color('none')
ax.xaxis.set_ticks_position('bottom')
ax.yaxis.set_ticks_position('left')

ax.minorticks_on()
ax.tick_params(axis='both', direction='inout', length=12, width=2, which='major')
ax.tick_params(axis='both', direction='inout', length=10, width=1, which='minor')
plt.show()
```



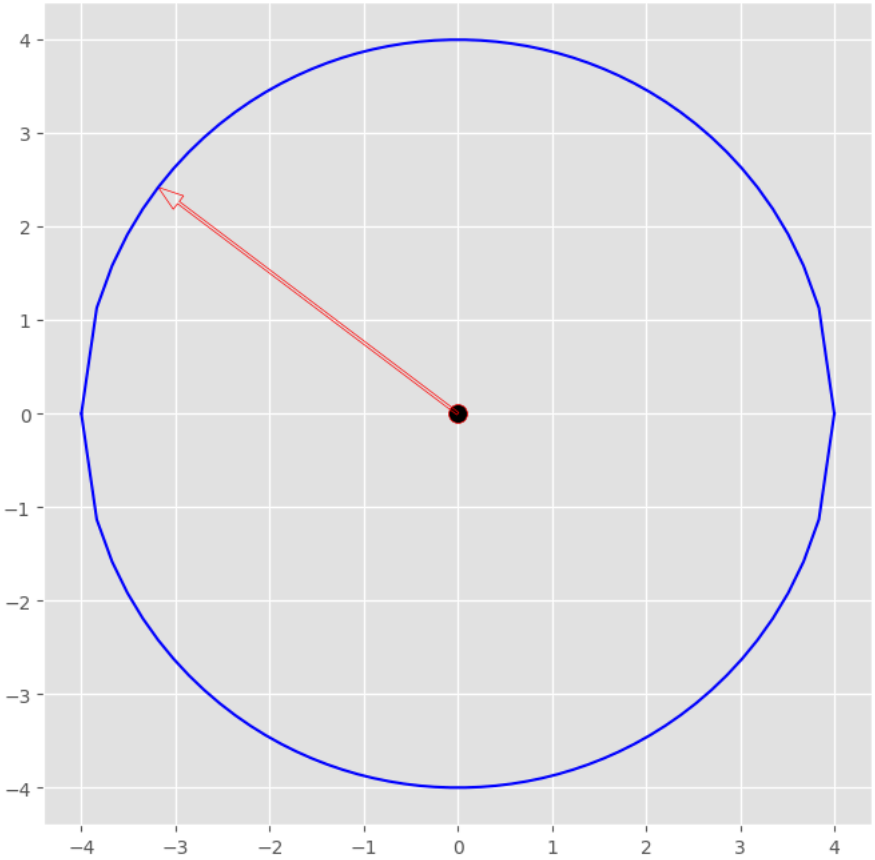
Geometric Intuition

Eigenvector has a special property that preserves the pointing direction after linear transformation.To illustrate the idea, let's plot a 'circle' and arrows touching edges of circle.

Start from one arrow. If you want to draw a smoother circle, you can use parametric function rather two quadratic functions, because cicle can't be draw with one-to-one mapping.But this is not the main issue, we will live with that.

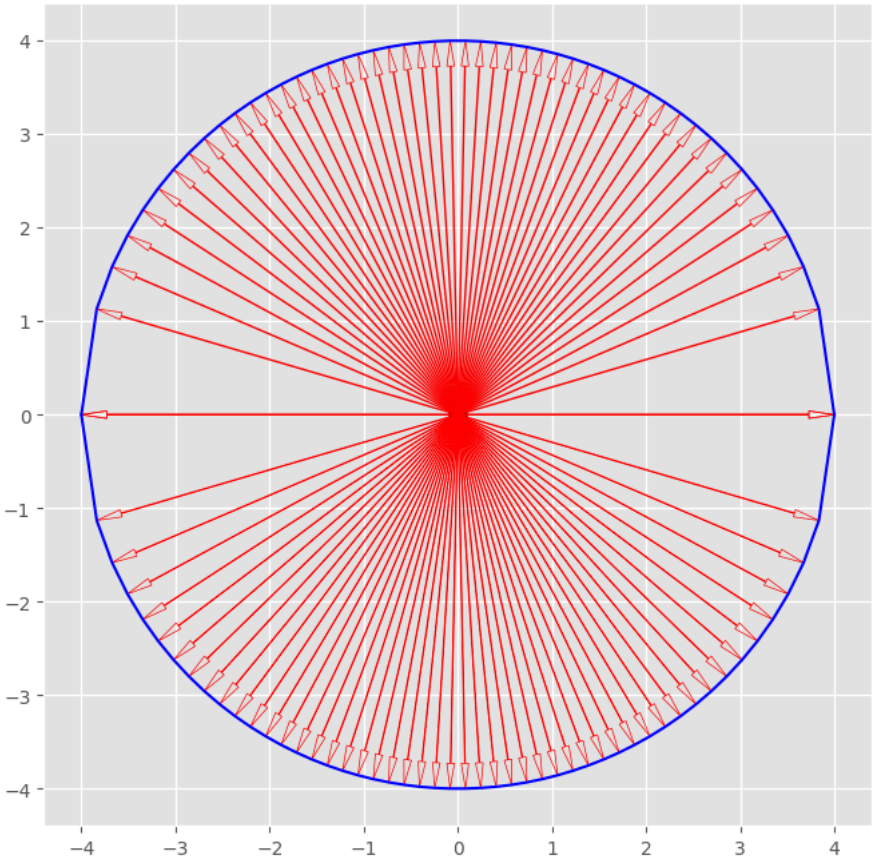
```
x = np.linspace(-4, 4)
y_u = np.sqrt(16 - x**2)
y_d = -np.sqrt(16 - x**2)

fig, ax = plt.subplots(figsize = (8, 8))
ax.plot(x, y_u, color = 'b')
ax.plot(x, y_d, color = 'b')

ax.scatter(0, 0, s = 100, fc = 'k', ec = 'r')

ax.arrow(0, 0, x[5], y_u[5], head_width = .18,
         head_length= .27, length_includes_head = True,
```

```
            width = .03, ec = 'r', fc = 'None')
plt.show()
```



Now, the same 'circle', but more arrows.

```
x = np.linspace(-4, 4, 50)
y_u = np.sqrt(16 - x**2)
y_d = -np.sqrt(16 - x**2)

fig, ax = plt.subplots(figsize = (8, 8))
ax.plot(x, y_u, color = 'b')
ax.plot(x, y_d, color = 'b')

ax.scatter(0, 0, s = 100, fc = 'k', ec = 'r')

for i in range(len(x)):
    ax.arrow(0, 0, x[i], y_u[i], head_width = .08,
             head_length= .27, length_includes_head = True,
             width = .01, ec = 'r', fc = 'None')
    ax.arrow(0, 0, x[i], y_d[i], head_width = .08,
             head_length= .27, length_includes_head = True,
             width = .008, ec = 'r', fc = 'None')
```



Now we will perform linear transformation on the circle. Technically, we can only transform the points - the arrow tip - that we specify on the circle.

We create a matrix

$$A = \begin{bmatrix} 3 & -2 \\ 1 & 0 \end{bmatrix}$$

Align all the coordinates into two matrices for upper and lower half respectively.

$$V_u = \begin{bmatrix} x_1^u & x_2^u & \dots & x_m^u \\ y_1^u & y_2^u & \dots & y_m^u \end{bmatrix}$$
$$V_d = \begin{bmatrix} x_1^d & x_2^d & \dots & x_m^d \\ y_1^d & y_2^d & \dots & y_m^d \end{bmatrix}$$

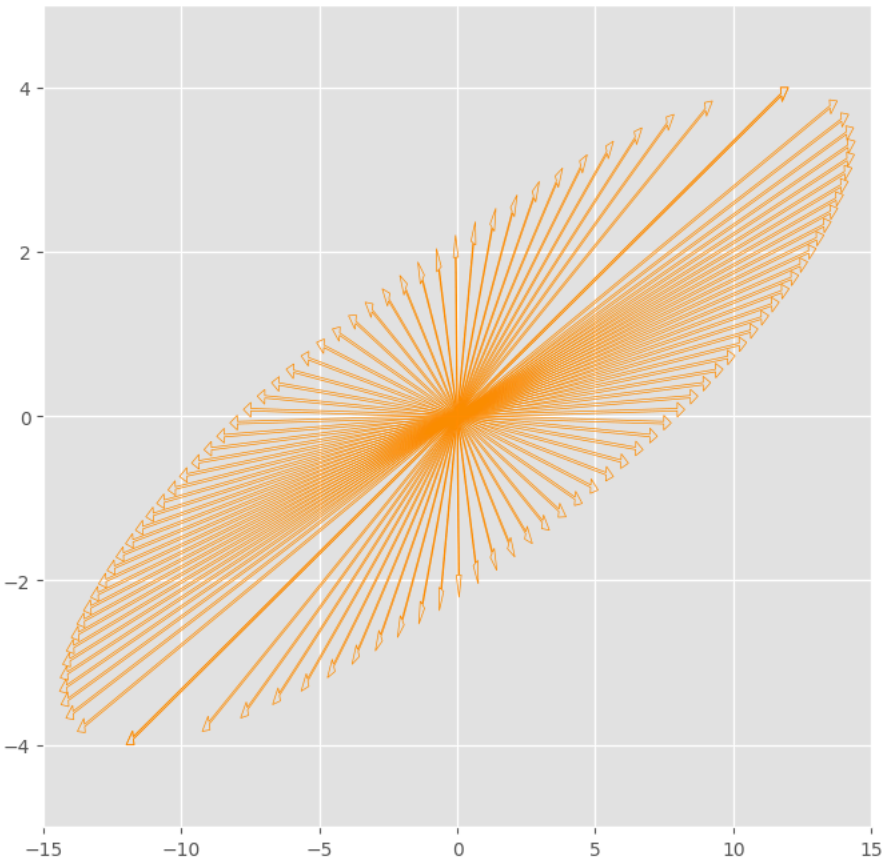
The matrix multiplication AV_u and AV_d are linear transformation of the circle.

```
A = np.array([[3, -2], [1, 0]])

Vu = np.hstack((x[:, np.newaxis], y_u[:, np.newaxis]))
AV_u = (A@Vu.T)

Vd = np.hstack((x[:, np.newaxis], y_d[:, np.newaxis]))
AV_d = (A@Vd.T)
```

The circle becomes an ellipse. However, if you watch closely, you will find there are some arrows still pointing the same direction after linear transformation.

```
fig, ax = plt.subplots(figsize = (8, 8))

for i in range(len(x)):
    ax.arrow(0, 0, AV_u[0, i], AV_u[1, i], head_width = .18,
             head_length= .27, length_includes_head = True,
```

```
            width = .03, ec = 'darkorange', fc = 'None')
    ax.arrow(0, 0, AV_d[0, i], AV_d[1, i], head_width = .18,
             head_length= .27, length_includes_head = True,
             width = .03, ec = 'darkorange', fc = 'None')
ax.axis([-15, 15, -5, 5])
plt.show()
```



We can plot the cirle and ellipse together, those vectors pointing the same direction before and after the linear transformation are eigenvector of A, eigenvalue is the length ratio between them.

```
k = 50
x = np.linspace(-4, 4, k)
y_u = np.sqrt(16 - x**2)
y_d = -np.sqrt(16 - x**2)

fig, ax = plt.subplots(figsize = (16, 8))

ax.scatter(0, 0, s = 100, fc = 'k', ec = 'r')

for i in range(len(x)):
    ax.arrow(0, 0, x[i], y_u[i], head_width = .18,
             head_length= .27, length_includes_head = True,
             width = .03, ec = 'r', alpha = .5, fc = 'None')
    ax.arrow(0, 0, x[i], y_d[i], head_width = .18,
             head_length= .27, length_includes_head = True,
             width = .03, ec = 'r', alpha = .5, fc = 'None')

A = np.array([[3, -2], [1, 0]])

v = np.hstack((x[:, np.newaxis], y_u[:, np.newaxis]))
Av_1 = (A@v.T)

v = np.hstack((x[:, np.newaxis], y_d[:, np.newaxis]))
Av_2 = (A@v.T)

for i in range(len(x)):
    ax.arrow(0, 0, Av_1[0, i], Av_1[1, i], head_width = .18,
             head_length= .27, length_includes_head = True,
             width = .03, ec = 'darkorange', fc = 'None')
```

```
    ax.arrow(0, 0, Av_2[0, i], Av_2[1, i], head_width = .18,
             head_length= .27, length_includes_head = True,
             width = .03, ec = 'darkorange', fc = 'None')
n = 7
ax.axis([-2*n, 2*n, -n, n])
plt.show()
```